

1 Categories

1.1 Data

^{1.1.1} A category $\mathbf{C}$ consists of the following data:

1. A collection $\text{Ob}(\mathbf{C})$ of objects.
2. For each pair of objects $X, Y \in \text{Ob}(\mathbf{C})$, a set $\text{Hom}_{\mathbf{C}}(X, Y)$ of morphisms from X to Y .
3. For each object X , an identity morphism $\text{id}_X \in \text{Hom}_{\mathbf{C}}(X, X)$.
4. For each triple of objects X, Y, Z , a composition map

$$\circ: \text{Hom}(Y, Z) \times \text{Hom}(X, Y) \rightarrow \text{Hom}(X, Z). \quad (1)$$

1.2 Axioms

^{1.2.1} subject to the following axioms:

Definition 1.2.2 (Category). *A category $\mathbf{C}$ consists of the data above, satisfying associativity and identity laws:*

$$h \circ (g \circ f) = (h \circ g) \circ f \quad (2)$$

and

$$f \circ \text{id}_X = f = \text{id}_Y \circ f.$$

Proposition 1.2.3. *Every category has a unique identity morphism for each object.*

^{1.2.4} *Proof.* Suppose e and e' are both identities for X . Then

$$e = e' \circ e = e'.$$

The first equality uses e' as a left identity; the second uses e as a right identity. $\square$

^{1.2.5} Another paragraph after the proof.

Remark 1.2.6. The identity morphism is unique by the argument above. This is a standard trick in algebra.

1.3 Examples

Example 1.3.1. The category **Set** has sets as objects and functions as morphisms. Composition is ordinary function composition:

$$(g \circ f)(x) = g(f(x)). \quad (3)$$

Example 1.3.2. The category **Grp** has groups as objects and group homomorphisms as morphisms.

^{1.3.3} A paragraph between examples.

Example 1.3.4. Any partially ordered set $(P, \leq)$ defines a category: objects are elements of P , and there is a unique morphism $x \rightarrow y$ iff $x \leq y$. Composition and identities are forced.

2 Functors

2.1 Definition

^{2.1.1} A functor maps one category to another, preserving composition.

Definition 2.1.2 (Functor). A functor $F: \mathbf{C} \rightarrow \mathbf{D}$ consists of:

1. An object map $F: \text{Ob}(\mathbf{C}) \rightarrow \text{Ob}(\mathbf{D})$.
2. A morphism map $F_{X,Y}: \text{Hom}_{\mathbf{C}}(X, Y) \rightarrow \text{Hom}_{\mathbf{D}}(F(X), F(Y))$.

satisfying:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad (4)$$

and

$$F(g \circ f) = F(g) \circ F(f). \quad (5)$$

2.2 Properties

Lemma 2.2.1. Functors preserve isomorphisms.

^{2.2.2} *Proof.* If $f: X \rightarrow Y$ is an isomorphism with inverse f^{-1} , then

$$F(f) \circ F(f^{-1}) = F(f \circ f^{-1}) = F(\text{id}_Y) = \text{id}_{F(Y)}.$$

Similarly $F(f^{-1}) \circ F(f) = \text{id}_{F(X)}$. □

Corollary 2.2.3. If $\mathbf{C}$ and $\mathbf{D}$ are isomorphic categories, then a functor $F: \mathbf{C} \rightarrow \mathbf{D}$ that is an equivalence preserves all categorical structure up to isomorphism.

^{2.2.4} A final paragraph. Cross-reference test: see Definition ?? if we had labeled it.